

Project Report: California House Price Prediction using Linear Regression

1. Introduction

This project aimed to develop and evaluate a Linear Regression model for predicting median house values in California using the California Housing dataset. The process followed a standard machine learning workflow, encompassing data loading, exploratory data analysis (EDA), data preprocessing, model training, evaluation, and a demonstration of its predictive capability.

2. Data Loading and Initial Exploration

- The California Housing dataset was loaded using `fetch_california_housing` from `sklearn.datasets`.
- The dataset contains **20,640 instances** with **8 numeric features** and one target variable: `MedHouseVal` (Median House Value).
- The target variable represents median house values in hundreds of thousands of dollars (\$100,000).
- The first 5 rows of the DataFrame were displayed to get an initial look at the data structure.

3. Exploratory Data Analysis (EDA) & Preprocessing

- **Missing Values and Data Types:**
 - `df.info()` and `df.isnull().sum()` were used to check for missing values and data types.
 - **Outcome:** No missing values were found across any features or the target variable. All features and the target were confirmed to be of `float64` data type, which is suitable for numerical analysis with Linear Regression.
- **Descriptive Statistics:**
 - `df.describe()` provided a statistical summary of the dataset, including count, mean, standard deviation, min, max, and quartiles for each numerical column. This helped in understanding the distribution and range of each feature.

4. Data Splitting

- **Feature and Target Separation:**
 - The features (X) were separated from the target variable (y), where y is MedHouseVal.
- **Train-Test Split:**
 - The dataset was split into training and testing sets to evaluate the model's performance on unseen data.
 - An **80% training set** and **20% testing set** split was used, with random_state=42 for reproducibility.
 - **Outcome:**
 - X_train shape: (16512, 8)
 - X_test shape: (4128, 8)
 - y_train shape: (16512,)
 - y_test shape: (4128,)

5. Model Training

- **Model Initialization:**
 - A LinearRegression model from scikit-learn was initialized.
- **Model Training:**
 - The model was trained using the X_train (features) and y_train (target) data.
 - **Outcome:** The Linear Regression model was successfully trained on the provided data.

6. Model Evaluation

- **Predictions:**
 - Predictions (y_pred) were generated on the X_test set.
- **Evaluation Metrics:**
 - The model's performance was assessed using standard regression metrics:

- **Mean Absolute Error (MAE): 0.53** (This implies an average prediction error of
- **R-squared (R^2): 0.58** (The model explains approximately 58% of the variance in median housing prices, indicating a moderate fit).
- **Visual Assessment:**
 - A scatter plot of actual vs. predicted values was generated. A red dashed line representing perfect prediction was included for visual comparison. This plot helps to identify any systematic biases or patterns in the model's predictions.

7. Model Demonstration

- **predict_house_value Function:**
 - A Python function predict_house_value was created to enable users to input specific feature values and receive a predicted median house value in dollars.
- **Example Usage:**
 - The function was demonstrated using the **mean values** of the dataset's features as input.
 - **Outcome:** For the average feature values, the model predicted a median house value of **\$206,786.23**.

8. Unresolved Questions/Tasks and Suggested Improvements

While the current Linear Regression model provides a baseline for house price prediction, there are several avenues for further improvement and task completion:

- **Further Model Improvements:**
 - **Feature Engineering:** Create new features from existing ones (e.g., polynomial features, interaction terms) to capture more complex relationships.
 - **Exploring More Advanced Models:** Investigate other regression models such as Ridge, Lasso, ElasticNet, Random Forests, Gradient Boosting Machines (e.g., XGBoost, LightGBM), or Neural Networks.
 - **Hyperparameter Tuning:** Optimize the hyperparameters of the chosen model(s) using techniques like GridSearchCV or RandomizedSearchCV.
 - **Feature Scaling:** Apply feature scaling (e.g., StandardScaler, MinMaxScaler) to features, which can improve the performance of many machine learning

algorithms, especially those sensitive to feature magnitudes (like Linear Regression with regularization).

- **Outlier Detection and Handling:** Identify and address outliers in the dataset, which can disproportionately influence model training.
- **Deliverables:**
 - **Jupyter Notebook:** The notebook containing all the code and plots is in progress.
 - **PDF Report:** A comprehensive 2-4 page PDF report summarizing the EDA, model details, evaluation metrics, and improvement ideas still needs to be finalized.
 - **Small UI Script:** The `predict_house_value` function partially addresses this, but a more comprehensive interactive UI could be developed.

Conclusion

The Linear Regression model successfully established a baseline for predicting California house prices, achieving an R-squared of 0.58 and an RMSE of \$75,000. While these results indicate a moderate fit, significant potential exists for enhancing model performance through further advanced techniques and thorough investigation of the dataset.